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Dataset exploring the atomic scale structure and ionic dynamics of polyanion sodium cathode materials

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Polyanionic sodium cathode materials exhibit promising electrochemical properties and high stability, making this chemical space worth exploring to enhance the performance of sodium-ion batteries. Given the vast chemical space, fast and efficient computational methods are needed, highlighting the need for machine learning (ML)-guided approaches. To support the ML-driven discovery, we have developed a comprehensive theoretical dataset comprising 113532 DFT-calculated structures with atomic charges and 184612 structures without atomic charges for four polyanion sodium-ion cathode materials: $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), Na_2TMSiO_4, and $Na_{2.56}TM_{1.72}(SO_4)_3$, with transition metals TM ∈ {Fe, Mn, Co, Ni}. This dataset includes optimizations of structures, ab initio molecular dynamics simulation trajectories sampled at 1000 K, and structures generated from ML-driven molecular dynamics simulation at 1000 K using an active learning algorithm. The dataset contains both single and multiple-TM compositions, providing a diverse representation of cathode materials. Using cathode specific subsets of this dataset to train ML models, we demonstrate that we can create ML interatomic potentials that reproduces the DFT results of the polyanion sodium cathode materials.

Background & Summary

The development of experimental[1–3] and theoretical databases[4–9] has become a primary focus in data-driven materials science in recent years[10]. The Materials Project (MP) database[11] has gained significant popularity in solid-state materials research and has facilitated the creation of several universal force fields[12–14], which are widely used to predict the physical properties of various materials[15,16]. However, the MP database does not encompass all compounds and primarily contains structures optimized to the lowest energy (i.e., ground-state structures). Efforts are underway to create databases that encompass a broader spectrum of compounds, which include site defects and account for ionic dynamics[17,18]. These expanded datasets are essential for building force fields capable of accurately predicting materials properties when the structures deviate from their ground state[15,19,20]. To further improve these models, system specific databases are needed, particularly for functional materials like battery electrodes, where capturing structures away from equilibrium can provide deeper insights and drive material performance improvements.

Sodium-ion batteries are considered a sustainable alternative to lithium-ion batteries, but they currently fall short in terms of performance. This limitation arises from the larger size of sodium (Na) compared to lithium (Li), despite both having the same number of valence electrons, leading to lower ionic conductivity[21,22]. Development of materials with higher ionic conductivity, particularly for cathode materials[23], is necessary to improve the performance of sodium-ion batteries. Among the various types of cathode materials, polyanionic compounds (NaTMA, where TM represents transition metals and A denotes the polyanion group) exhibit the highest stability and offer good energy density.

A key limitation of the polyanionic compounds is their poor electronic transport properties[21,23]. One strategy to enhance electronic transport is doping, which involves incorporating multiple types of TM ions or A groups into the cathode material composition[24]. This approach can facilitate the electron transport, which is essential for the redox processes that occur during Na-ion extraction[25]. However, the phase space of potential dopants for

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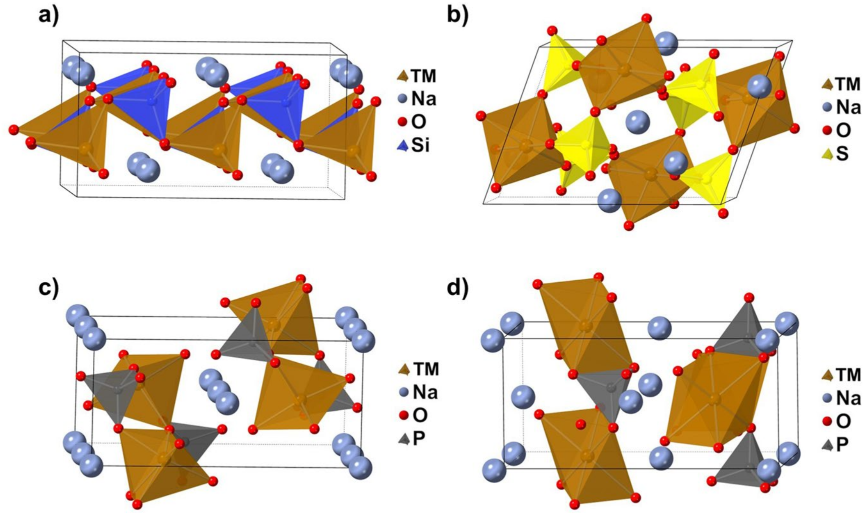

Fig. 1 The phase space of the polyanion sodium cathode materials dataset. The structures depict a unit cell of (**a**) Na_2TMSiO_4 with a total of 32 atoms, (**b**) $Na_{2.56}TM_{1.72}(SO_4)_3$ with a total of 77 atoms, (**c**) $NaTMPO_4$ (olivine) with a total of 28 atoms and (**d**) $NaTMPO_4$ (maricite) with a total of 28 atoms.

Na-ion polyanionic cathode materials is vast and remains largely unexplored in existing databases. This, coupled with the need to account for ionic conductivity, motivates the creation of the "polyanion sodium cathode materials dataset" presented in this work.

The materials in our dataset contains TM ions capable of undergoing the $TM^{2+} \rightleftharpoons TM^{3+}$ redox process. The dataset includes four types of compounds — $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), Na_2TMSiO_4, and $Na_{2.56}TM_{1.72}(SO_4)_3$ — which are among the most widely used polyanionic materials for Na-ion battery cathodes. All of these compounds incorporate one or more of the four TM ions, where TM $\in$ {Fe, Mn, Co, Ni}, and considers various degrees of sodiation and TM combinations.

To systematically explore the configurational space for each material, we enumerate all symmetrically inequivalent structures when their number is below 700; otherwise, we randomly sample different Na- and TM-ion arrangements when the number exceeds this threshold, as full enumeration becomes computationally prohibitive. All sampled structures are optimized using density functional theory (DFT). From this set, additional structures are generated through ab initio molecular dynamics (AIMD) simulation at 1000 K. To further extend the simulation timescales, machine learning (ML)-driven molecular dynamics (MD) simulations are employed. By utilizing an active learning (AL) algorithm, the most informative structures along the trajectory are identified. Single-point DFT calculations are then performed for these structures and incorporated into the training set of the ML model to enhance its predictive accuracy.

Using this framework, we have constructed a dataset that includes both energy optimizations of structures and MD sampling for four NaTMA materials, along with various structures incorporating TM ion doping. For all sampled structure, we record its crystal composition, total energy, atom-wise force vectors and atom-wise magnetic moments derived from Mulliken analysis[26]. For all the lowest energy structures and MD sampled configurations, we record the atomic charges obtained through Bader charge analysis[27]. Our polyanion sodium cathode materials dataset serves as a valuable addition to existing datasets, facilitating phase space exploration while providing insights into the dynamic behavior of these materials.

Methods

The phase space of the polyanion sodium cathode materials dataset is illustrated in Fig. 1. We classify the dataset into two main categories depending on the number of TM ion types present in the structure: single TM ion structures, which contain only one TM type, and multiple TM ion structures, which incorporate more than one TM type. Low-energy structures are sampled for both categories, whereas dynamic behavior is sampled exclusively for the single TM ion category.

The unit cell structures for $NaTMPO_4$ (olivine) and $NaTMPO_4$ (maricite) are obtained from ref. [28], where the maricite structure is generated by swapping the positions of the Na and TM ions in the olivine structure[23]. The unit cell structures for Na_2TMSiO_4 and $Na_{2.56}TM_{1.72}(SO_4)_3$ are taken from ref. [22] and ref. [29], respectively.

For $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), and Na_2TMSiO_4, structure generation follows a standardized workflow for both low-energy structures and AIMD sampling. In contrast, for $Na_{2.56}TM_{1.72}(SO_4)_3$, a random sampling approach is used due to its substitutional disorder, where Na and TM ions can swap positions. MD sampling is performed for the single TM ion structures for all four cathode materials using AIMD and ML-driven MD sampling methods.

Structure generation workflow. To systematically generate symmetrically inequivalent configurations for a given cathode material with varying Na concentrations, two degrees of freedom must be considered:

Atom	Redox potential ordering	TM^{2+}	TM^{3+}
Fe	First	$4\,\mu_B$	$5\mu_B$
Mn	Second	$5\mu_B$	$4\mu_B$
Co	Third	$3\mu_B$	$4\mu_B$
Ni	Forth	$2\mu_B$	$3\mu_B$

Table 1. Redox potential ordering and magnetic moment for the different transition metal (TM) ions.

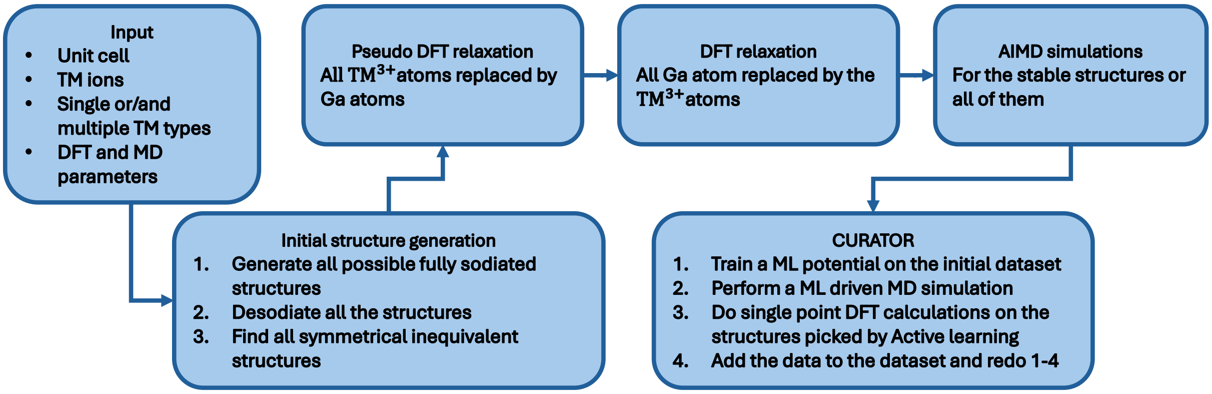

Fig. 2 Structure generation workflow. From the input file, a symmetrical enumeration of all possible structures is performed within a specified Na-ion cathode material with predetermined transition metal (TM) ions. The inequivalent generated structures undergoes a two-step density functional theory (DFT) relaxation to obtain the ferromagnetic state of the structures with preset oxidation state for the TM ions. Ab initio molecular dynamics (AIMD) simulations are then conducted using either all structures or only the most stable ones determined from the lowest energy. The combined results of the DFT and AIMD calculations form an initial dataset, which is utilized with the CURATOR[42] package to train a machine learning interatomic potentials, enabling ML-driven MD sampling.

(i) the positions of the Na ions within the structure and (ii) the oxidation state of the TM ions, which dictate the redox processes occurring when Na ions are added or removed — a concept known as configurational electronic entropy[30]. For materials with multiple TM ion types, a third degree of freedom arises from the spatial distribution of different TM ions.

In systems containing multiple TM ion types, redox processes preferentially occur at the atom with the lowest redox potential. Additionally, as TM ions undergo redox transition from $TM^{2+} \rightarrow TM^{3+}$, their spin states also change. The ordering of redox-active TM ions and their associated magnetic moments are summarized in Table 1. All generated structures are fixed to a ferromagnetic ordering, as exploring all the possible antiferromagnetic configurations would increase the search space exponentially.

Figure 2 illustrates the structure generation workflow, implemented using the PerQueue workflow manager[31]. The workflow starts by loading the unit cell structure of a cathode material containing a single TM ion type. Subsequently, users can choose to sample structures with either single TM ion types and/or multiple TM ion types. The parameters for the DFT and AIMD calculations are provided in the input file.

Initially, all possible TM ion arrangements for both single TM and multiple TM ion structures are generated in the fully sodiated state. The desodiation process then proceeds systematically, removing Na ions in a combinatorial manner. In the desodiated states, Ga atoms are used as a marker to indicate which TM ions undergo the redox processes, as described in Table 1. We enumerate all possible configurations to consider all possible the redox processes that can happen at each level of sodiation. To ensure that only symmetrically inequivalent structures are retained, the Atomic Simulation Environment (ASE)[32] symmetry equivalence tool is used to filter the redundant structures.

When Na ions are removed, targeted TM ions must be oxidized (i.e., transition from TM^{2+} to TM^{3+}). To ensure that oxidation occurs at specific TM sites, a "pseudo-relaxation" step is performed before the actual relaxation. In this step, oxidizing TM ions are temporarily substituted with Ga ions, which preferentially adopt a stable +3 oxidation state (oxidation state of +2 is rather unstable for Ga) and have an ionic radius similar to the TM ions considered. This ensures that when n Na ions are removed, exactly n Ga maintain +3 oxidation state, while the remaining TM ion retain their +2 oxidation state. During this pseudo-relaxation, the total magnetic moment is fixed, TM ions are assigned predefined magnetic moments according to Table 1, and the magnetic moment of Ga ions is set to $0\mu_B$. Once this step converges, the Ga ions are replaced by the original TM ions, followed by a final relaxation with the total magnetic moment of the system fixed, following the same scheme of ref. [30]. This two-step process, developed by ref. [33], ensures that oxidation occurs at predefined positions and that the structures retain ferromagnetic ordering, as demonstrated in ref. [34].

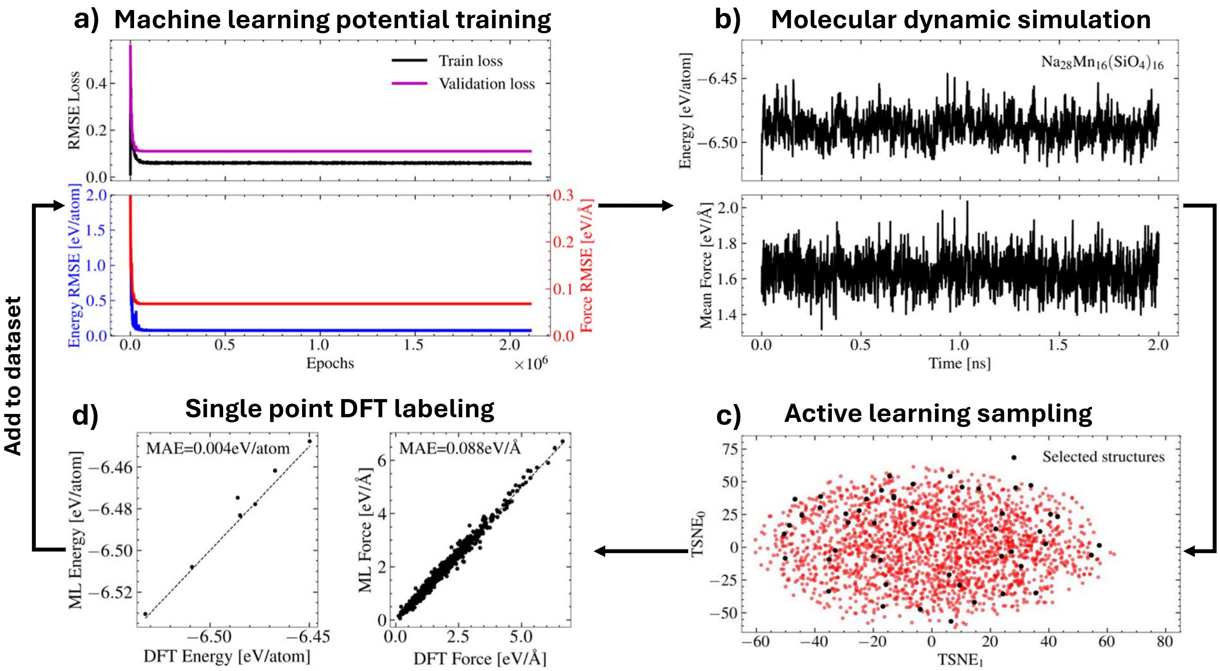

Fig. 3 The CURATOR[42] framework utilized to train machine learning interatomic potentials (MLIPs) for a specific cathode phase space, exemplified by the second iteration of the Na_2TMSiO_4 material for the $Na_{28}Mn_{16}(SiO_4)_{16}$ structure. (**a**) The initial dataset, along with prior iteration data, is used to train the Polarizable Atom Interaction Neural Network (PaiNN)[36] with 128 hidden nodes and 4 interaction layers. The loss metric is root mean square error(RMSE). (**b**) The trained PaiNN model is employed to perform molecular dynamics (MD) simulations to explore the dynamical behavior of Na ions. (**c**) Batch Active Learning (BAL)[39] selects a diverse and informative sample set from the ML-driven MD simulation, with t-SNE[55] visualizing the feature space in 2D. (**d**) The selected samples are labeled with Density Functional Theory (DFT) calculations, and the results are compared with MLIP predictions using the mean absolute error (MAE) metric. All converged DFT data are added to the expanding dataset, and the workflow continues until DFT and MLIP predictions align, enabling MD simulations up to 2 ns.

After the structure generation, AIMD simulations are performed on either all low-energy structures or only the most stable configurations for each sodiation level, with the magnetic moments still fixed. Stability are determined based on the structure with the lowest energy for each unique TM ion composition at a given sodiation level. The structures sampled during optimization, the low-energy structures and AIMD simulation trajectories are then compiled into an initial database, which is then used to train a machine learning interatomic potential (MLIP) for ML-driven MD sampling.

The structure generation workflow is applied to $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite) and Na_2TMSiO_4 to obtain both low-energy structures and AIMD trajectories for the single TM ion structures, as well as low-energy structures for the multiple TM ion structures. A symmetry tolerance of 1e-3 and a distortion distance of 0.004 Å is used for all structure generations.

The structure generation workflow, along with the scripts for dataset generation, is provided in the Code Availability section. This methodology is also applicable to other intercalation cathode materials like those incorporating Li or Mg as mobile ions.

Random sampling. For materials exhibiting substitutional disorder, where Na and TM ions can swap positions, enumerating all the possible symmetrically inequivalent structures for a given composition becomes infeasible. To address this, we employ the CLEASE package[35] to incorporate the disorder into the $Na_{2.56}TM_{1.72}(SO_4)_3$ structure search space. Using the random structure sampler, several configurations are selected and optimized with DFT to obtain the low-energy structure. During the DFT optimization the magnetic moments are kept fixed.

AIMD simulations are then performed on the thermodynamically stable configurations at each sodiation level, determined by the structures with the lowest energy. The structures sampled during optimization, the low-energy structures and AIMD simulation trajectories are incorporated into the initial dataset for the MLIP.

For the multiple TM ion structures, possible configurations are generated by doping additional TM ion types into the low-energy single TM ion structures. From this set, an AL algorithm is employed to select a subset of the generated pair TM ion structures for structural optimization. This process results in a dataset identified by the ML model as the most significant and diverse.

Machine learning-driven molecular dynamic sampling. Using an initial dataset generated from either the structure generation workflow or the random sampling, we train the polarizable atom interaction neural network (PaiNN) model[36]. The PaiNN model efficiently predicts energy and forces with an accuracy comparable to DFT[37]. We extend the timescales beyond those accessible by AIMD by leveraging ML-driven MD, allowing for the extraction of dynamical properties such as the diffusion coefficient of Na ions[38]. This enables the ML model to extrapolate predictions to previously unencountered atomic environments.

Rather than randomly sample unknown environments, we utilize Batch Active Learning (BAL)[39]. Unlike naive active learning, where independently selected samples (within a batch) often exhibit high similarity, BAL simultaneously selects samples while prioritizing both uncertainty and diversity within the batch[40]. The BAL algorithm first constructs the full gradient kernel from all features within the ML model. This kernel is then transformed using random projection, followed by greedy maximization in the largest clusters (greedy-LCMD)[39] to cluster data points in the feature space. Each new cluster center is determined by maximizing the distance between the other cluster centers, using Mahalanobis distance as the metric[41]. By selecting only the cluster centers as sample points, this deterministic method ensures a diverse and informative sample set.

Following BAL, single-point DFT calculations are conducted to relabel the selected samples. The converged structures are incorporated into the initial dataset, which is then used to retrain the MLIP. This iterative process is illustrated with an example case in Fig. 3. This workflow is implemented using CURATOR[42], which utilizes MyQueue[43] to autonomously perform all workflow steps in parallel for each structure in the material group under consideration.

CURATOR is employed to execute ML-driven MD sampling for all four cathode materials independently, resulting in four cathode material specific MLIPs. To facilitate better sampling of Na-ion diffusion, the unit cells used in AIMD are expanded into supercells for the ML-driven MD sampling. A $(1 \times 2 \times 2)$ supercell is used for $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite) and Na_2TMSiO_4, while a $(1 \times 1 \times 2)$ supercell is used for $Na_{2.56}TM_{1.72}(SO_4)_3$ as it has a larger unit cell as vizualized in Fig. 1. The criteria for ML-driven MD sampling include achieving a stable MD simulation with a total duration of 2 ns for all structures within the CURATOR framework, and ensuring sufficient MLIP prediction accuracy.

Within CURATOR, an ensemble of PaiNN models are used to statistically estimate uncertainties in forces and energies, ensuring control over the ML-driven MD simulations. For each CURATOR framework, six different PaiNN models are trained, each with different configurations: hidden node size of 120, 124 or 128, while the number of interaction layer is either 3 or 4. All models are configured with a message passing cutoff of 4 Å, accounting for interaction between atoms within the cutoff, and a batch size of 16. All models employ atom-wise energy normalization, and the total loss function is a weighted sum of the mean absolute error (MAE) for energy and force, with the force loss weighted at 0.98 and the energy loss at 0.02. The ADAM optimizer[44] with an initial learning rate of 0.001, which is dynamically adjusted via a learning rate scheduler when no significant improvement in loss is detected.

Density functional theory. All DFT calculations are performed using the Vienna ab initio simulation package (VASP)[45] version 6.4, which employs the projected-augmented wave (PAW)[46] to describe electron-ion interactions. The Perdew-Burke-Ernzerhof (PBE) functional[47] is used for all calculations. To mitigate electronic self-interaction error, which can significantly affect systems with strong localization of *d*-orbital electrons, Hubbard-*U* corrections are applied within the Kohn-Sham equation to account for the electronic localization of the 3*d* orbitals of the TM ions (TM ∈ {Fe, Mn, Co, Ni})[28,48]. The *U* values for the TM ions are selected from experimentally verified structures in the Materials Project database[49], resulting in the following assignments: *U*= (Fe: 5.3 eV, Mn: 3.9 eV, Co: 3.32 eV, Ni: 6.2 eV)[50]. For all calculations, an energy cutoff of 520 eV is applied, with a smearing width of 0.01 eV and convergence criteria set to 1×10^{-5} eV for the energy and 0.03 eV/Å for the forces. All calculations are performed with spin polarization. The *k*-point grids employed for the four cathode materials are fixed, with $NaTMPO_4$ (olivine) and $NaTMPO_4$ (maricite) utilizing $(3 \times 4 \times 6)$ grids, Na_2TMSiO_4 employing $(3 \times 4 \times 4)$ grids and $Na_{2.56}TM_{1.72}(SO_4)_3$ utilizing $(2 \times 3 \times 4)$ grids. In all cases, the *k*-point grids were γ-centered. When constructing supercells, the *k*-point grids was reduced by half in the in the direction for which the unit cell was enlarged. Atomic charges are obtained by performing Bader charge analysis on the saved CHGCAR files containing the charge densities[27].

Molecular dynamics simulation. All MD simulations are conducted using the Langevin thermostat[51] with a friction constant of 0.003. The temperature is maintained at 1000 K to promote diffusion events, and a time step of 1 fs is used throughout. All simulations are conducted within the canonical (NVT) ensemble[52]. All AIMD simulations ran for 48 hours of computational time on a single node in a high-performance computing cluster (40 CPU cores on Intel Xeon Gold 6148 with 384GB of RAM memory), resulting in different simulation lengths.

For AIMD simulations, structures are recorded at each time step. In ML-driven MD simulations, structures are saved at 10 fs intervals for simulations that last less than 1 ns and at 100 fs intervals for simulations exceeding 1 ns.

Data Records

The polyanion sodium cathode materials dataset, formatted in XYZ format, is available for downloaded at Zenodo[53]. The dataset consists of four sodium-ion battery cathode materials — $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), Na_2TMSiO_4 and $Na_{2.56T}TM_{1.72}(SO_4)_3$ — where TM ∈ {Fe, Mn, Co, Ni}. All data is formatted in XYZ files, allowing for a simple extraction of atomic structures and physical properties. Each dataset entry includes the total energy (in eV), force per atom (in eV/Å), stress tensor (in GPa) and magnetic moment per atom (in μ_B). The atomic charge (in e) obtained via Bader charge analysis[27] is also included for all low-energy structures and MD-sampled structures. All structures included in the dataset have electronically converged. A Python script is provided to demonstrate the extraction of atomic structures and associated physical properties, available in the Code Availability section.

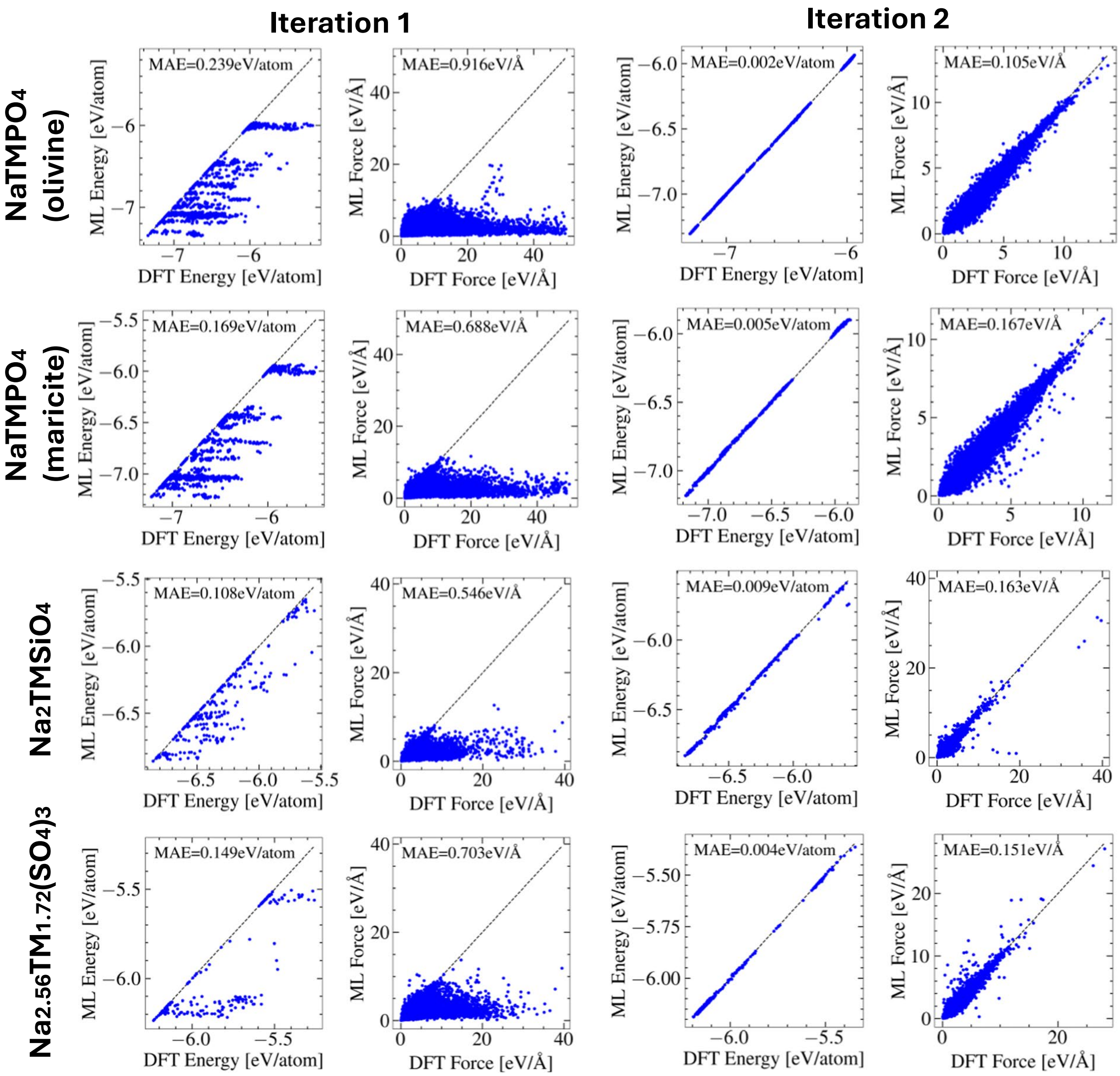

Fig. 4 The labeling layer of CURATOR[42] for each iteration of the four sodium-ion cathode materials. The energy and forces from single-point Density Functional Theory (DFT) calculations, selected using the Batch Active Learning (BAL) algorithm[39], are compared to their machine learning (ML) counterparts, using the mean absolute error (MAE) metric.

The dataset is divided into single TM ions structures and multiple TM ion structures, categorized separately for each of the four cathode materials: $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), Na_2TMSiO_4, and $Na_{2.56}TM_{1.72}(SO_4)_3$. For example, $Na_{2.56}TM_{1.72}(SO_4)_3$ structures are split into single TM ion structures `Na2M2SO4_alluadite_single_total.xyz` and multiple TM ion structures `Na2M2SO4_alluadite_multiple_optimization.xyz`. The single TM ion structures are further separated into the sampling methods used to gather the dataset: structural optimization `Na2M2SO4_alluadite_single_optimization.xyz`, AIMD simulation `Na2M2SO4_alluadite_single_AIMD.xyz`, and ML-driven MD sampling `Na2M2SO4_alluadite_single_MD_sampling.xyz`. This allows the user to either use the whole dataset or only structures gathered using a specific sampling method.

In total, 108267 single TM ion structures are sampled across all four cathode materials. In total, 5256 multiple TM ion structures are sampled across the four cathode materials. All this data includes atomic charges and is stored in the `polyanion_cathode_dataset` folder. The combined dataset, consisting of 113532 structures, is available at `polyanion_cathode_dataset/Combined.xyz`.

Beyond the 113532 structures with atomic charges, additional structures were obtained during the structural optimization process, which do not include atomic charges but retain all other physical properties. Thus, the dataset is divided into two parts: (1) a set of 113532 structures with atomic charges and (2) a larger dataset of 298144 structures containing all our data, including optimization steps. This additional dataset is stored in the

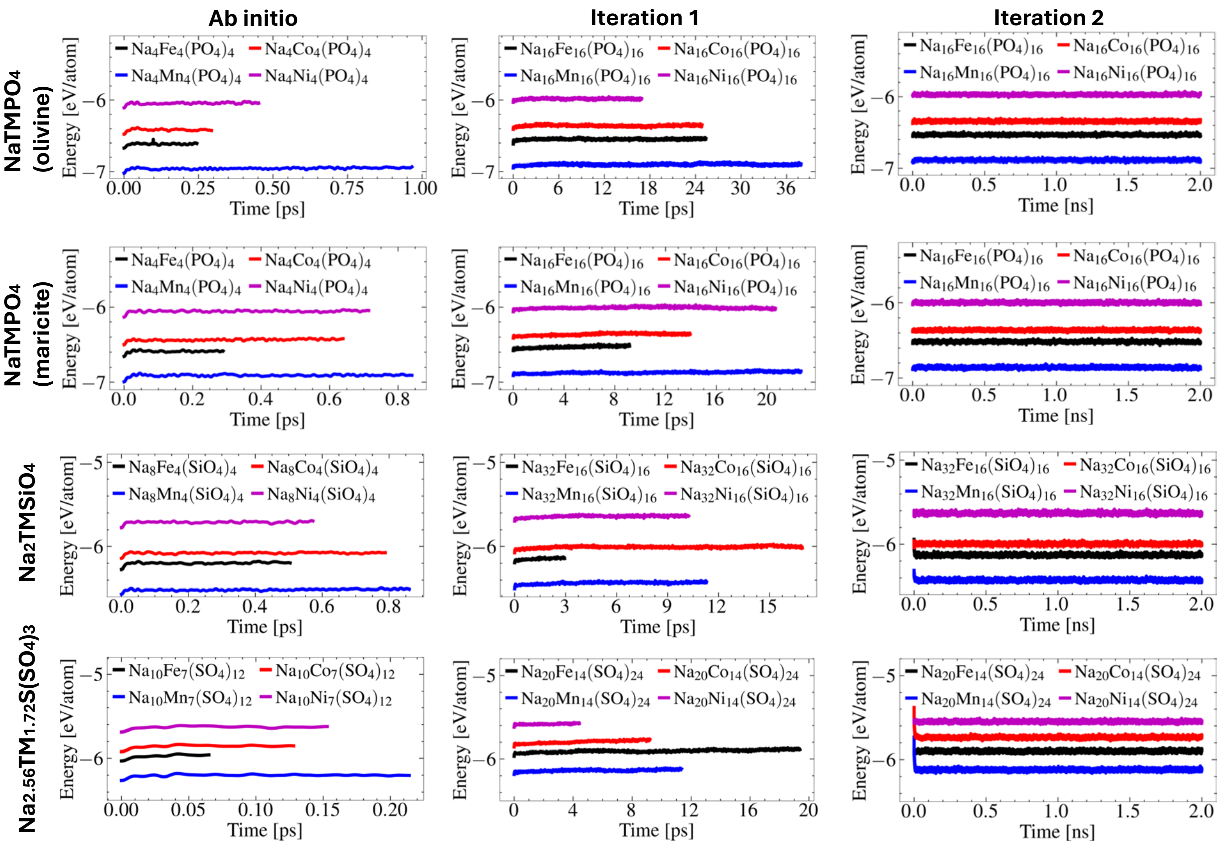

Fig. 5 Ab initio molecular dynamic (AIMD) simulations and machine learning (ML)-driven molecular dynamic (MD) simulations for iteration 1 and iteration 2 of the machine learning interatomic potential (MLIP) training for each of the four cathode materials. All simulation shows the total energy as a function of simulation time. The simulations are showed for each of the four sodium-ion cathode materials. In each material group four fully sodiated atomic structure compositions are simulated with their chemical formula presented and color coded.

`polyanion_cathode_dataset_with_optimization_steps` folder and is divided into single TM ions structures and multiple TM ion structures, categorized separately for each of the four cathode materials. For example, the structures without atomic charge from the structural optimization process of $Na_{2.56}TM_{1.72}(SO_4)_3$ single TM ion structures are collected in `polyanion_cathode_dataset_with_optimization_steps/Na2M2SO4_alluadite_single_structural_optimization.xyz`.

The complete dataset, including all structures with atomic charge as well as the structures obtained during structural optimization (298144 structures), is available at `polyanion_cathode_dataset_with_optimization_steps/Combined.xyz`.

Technical Validation

Figure 4 depicts the outcomes of each iteration in the ML-driven MD sampling for each of the four cathode materials MLIP training. In the first iteration when the MLIP are trained on the optimized structures and the AIMD simulations, the ML-predicted energy and forces significantly deviate from the single-point DFT calculations, found by using the BAL algorithm. This indicates that the ML model does not yet accurately capture the true values. However, by incorporating the single-point DFT calculations into the initial datasets and retraining the MLIP leads to a substantial improvement in the second iteration. This is clear from the new set of single-point DFT calculations found by using the BAL algorithm, where the energy MAE decreases tenfold and the force MAE decreases fivefold. The outcome demonstrates the efficiency of the BAL sampling in selecting the most informative structures, from the ML-driven MD simulations, for improving MLIP predictions, particularly in the context of high-temperature MD simulations.

Figure 5 highlights the importance of ML-driven MD sampling in capturing the phase space of cathode materials. Using 48 hours of computational time on a single node in a high-performance computing cluster (40 CPU cores on Intel Xeon Gold 6148 with 384GB of RAM memory), AIMD simulations are limited to sampling only 1 ps-2 ps per structure. While this duration is sufficient for reaching thermal equilibrium, a much longer simulation time is required to investigate diffusion processes[38,52,54]. Based on these results, achieving 2 ns of simulation solely through AIMD would require approximately 2000 days of computational time. However, by integrating AIMD simulations within the CURATOR workflow, ML-driven MD simulations can extend the accessible time scale by a factor of 10 to 20 before yielding unphysical results. Employing the BAL algorithm,

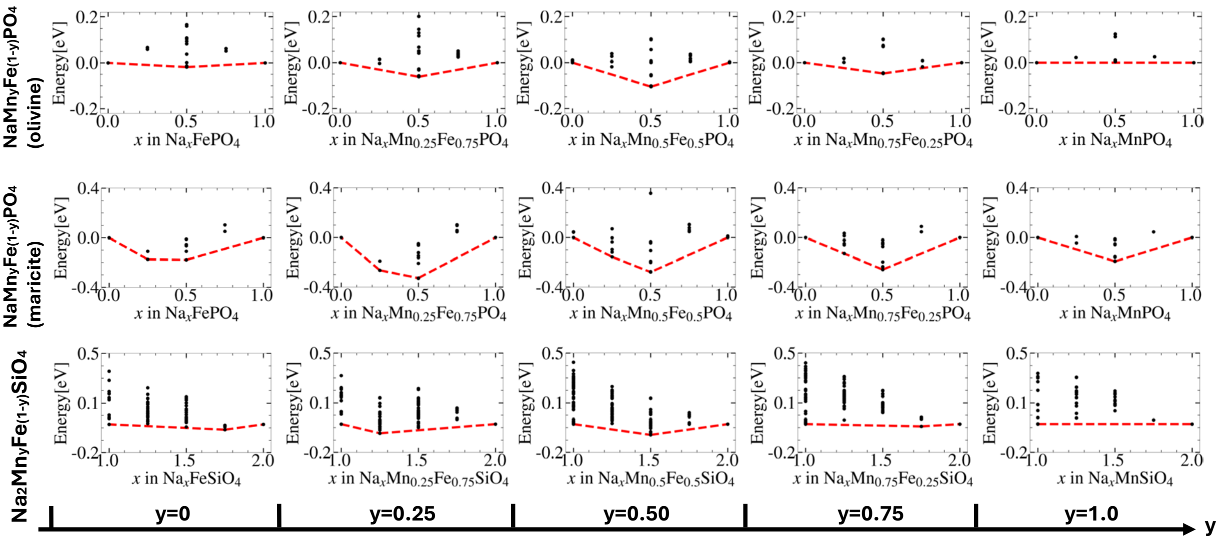

Fig. 6 The convex-hull of the ab initio calculation for the Fe-Mn pair TM ion types for the three materials, $Na_xMn_yFe_{1-y}PO_4$(olivine), $Na_xMn_yFe_{1-y}PO_4$(maricite), $Na_xMn_yFe_{1-y}SiO_4$ with x varying in each plot at different y. For each convex-hull plot. The energies are plotted relative to the reference points.

we select the most critical structures for single-point DFT calculations. Incorporated these into the training set, enabling retraining of the MLIP to improve prediction accuracy. By the second iteration, we achieve 2 ns of Na-ion dynamics simulation with accuracy comparable to DFT, as illustrated in Fig. 4. Notably, transitioning from AIMD to the second iteration requires significantly less time than the projected 2000 days. The final MLIP-based simulations require only 1-2 days of computational time on a single Nvidia RTX3090 GPU to complete a 2 ns MD simulation on the supercells of the materials. These ML-driven MD simulations, thus play a crucial role in our dataset, providing the necessary dynamical sampling of structure distribution to characterize the ionic conductivity of the cathode materials for different Na concentrations.

Figure 6 presents the convex hull for the Fe-Mn pair TM ion system, illustrating varying Na concentrations and different Fe/Mn ratios. This analysis is conducted for three materials: $NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite) and Na_2TMSiO_4, where pair TM ion type sampling is carried out for all possible symmetrical inequivalent structures using the structure generation workflow. The convex hull illustrates two reference points — the fully sodiated and fully desodiated states — corresponding to the charged and discharged states of a battery, respectively. The convex hull representation enables the calculation of open-circuit voltage[28] and provides insight into the charge/discharge behavior of cathode materials. By symmetrically enumeration of all symmetrically inequivalent configurations of the multiple TM ion type structures, we effectively model the disorder associated with TM and Na ions, providing essential descriptions needed for modeling larger structures utilizing ML methods or alternative techniques.

In total the polyanion sodium cathode material dataset contains 113532 structures with atomic charge and 184612 structures without atomic charge for four polyanion sodium cathode materials,$NaTMPO_4$ (olivine), $NaTMPO_4$ (maricite), Na_2TMSiO_4 and $Na_{2.56T}TM_{1.72}(SO_4)_3$ with transition mteal ions TM $\in$ {Fe, Mn, Co, Ni}. This dataset represents a significant contribution to the field, offering a comprehensive description of both low-energy structures and MD sampling, capturing the physical properties of polyanionic cathode materials across a broad phase space.

The MD sampling in this study (both AIMD and ML-driven MD) enables the characterization of ionic conductivity, while utilizing MLIPs allow for the analysis of Na-ion transport behavior in these cathode materials. Furthermore, incorporating atomic charge information into MLIPs can enable a direct description of the redox process of TM ions during battery charge/discharge cycles, a key factor in understanding ionic conductivity.

Code availability

The structure generation workflow and the random sampling approach are available in the cathode-generation-workflow repository (https://github.com/dtu-energy/cathode-generation-workflow) version 1.0. Detailed instructions for using the scripts to generate datasets can be found in the repository and accompanying README file. The CURATOR[42] version 1.0.0 is used to drive the ML-driven MD sampling.

To extract structural compositions and physical properties of the polyanion sodium cathode materials dataset, the `ase.io.read` function from ASE[32] version 3.23.0 is used. A Python script demonstrating how to extract data and plot the physical properties is provided in (https://github.com/dtu-energy/cathode-generation-workflow/tree/main/extract_data/).

All electronic structure calculations are conducted using ASE[32] version 3.23.0 and VASP[45] version 6.4, with the VASP license loaded independently. Random sampling is performed using CLEASE[35] version 1.0.6, and an example for the $Na_{2.56}TM_{1.72}(SO_4)_3$ material are provided in `Random_sampling/Random.ipynb` within the cathode-generation-workflow repository along with code to perform the structural optimization and AIMD. The structure generation workflow is managed using MyQueue[43] version 22.7.1 and PerQueue[31] version 0.2.2. These workflow managers are pre-configured for use on specific HPC clusters.

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Author contributions

A.B., J.H.C. and J.M.G.L. conceived and supervised this study; M.H.P. performed all the DFT simulations and trained the MLIPs; M.H.P. wrote the paper; A.B., J.M.G.L., J.H.C. reviewed and edited the paper. All authors have read and agreed to the current version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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